which is divisible by $a-b$. This implies that $P(k+1)$ is true. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction, $a^n - b^n$ is divisible by $a - b$, where $a > b$, for all natural numbers n.

Example 4.66 Prove that $3^{2n+2} - 8n - 9$ is divisible by 8 for all $n \geq 1$.

Solution:
Let

$$P(n) := 3^{2n+2} - 8n - 9, \text{ is divisible by } 8.$$

Substituting the value of $n = 1$, in the statement we get,

$$P(1) = 3^{2+2} - 8(1) - 9 = 64,$$

which is divisible by 8. Hence, $P(1)$ is true. Let us assume that the statement is true for $n = k$. Then $P(k) = 3^{2k+2} - 8k - 9$, is divisible by 8. We can write

$$\begin{aligned} P(k) &= 3^{2k+2} - 8k - 9 = 8k_1, \quad k_1 \in \mathbb{N} \text{ and therefore,} \\ 3^{2k+2} &= 8k_1 + 8k + 9. \end{aligned}$$

We need to show that $P(k+1) = 3^{2(k+1)+2} - 8(k+1) - 9$, is divisible by 8.
Consider,

$$\begin{aligned} P(k+1) &= 3^{2(k+1)+2} - 8(k+1) - 9 \\ &= 3^2 3^{2k+2} - 8k - 8 - 9 \\ &= 3^2 \left(8k_1 + 8k + 9\right) - 8k - 17 \\ &= 72k_1 + 64k + 64 \\ &= 8\left(9k_1 + 8k + 1\right) \\ &= 8k_2, \ k_2 = 9k_1 + 8k + 1 \in \mathbb{N} \end{aligned}$$

which is divisible by 8. This implies that $P(k+1)$ is true. This means that the validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction, $3^{2n+2} - 8n - 9$ is divisible by 8 for all $n \geq 1$.

Example 4.67 Using the Mathematical induction, show that for any integer

$$n \geq 2, \ 3n^2 > (n+1)^2$$

Solution:
Let $P(n)$ be the statement that $3n^2 > (n+1)^2$ with $n \geq 2$. Therefore the first stage is $n = 2$.
Now, $P(2) = 3 \times 2^2 = 12$ and $3^2 = 9$. As $12 > 9$ we get $P(2)$ is true.
We assume that $P(n)$ is true for $n = k$.